Pegah Karimi Ph.D.

Human Factors & UX Designer · UX Researcher

Usability engineering for clinical, safety-critical, and AI-driven products

Palo Alto, CA · pekarimi67@gmail.com · LinkedIn · Google Scholar · Portfolio

PROFILE

Human factors and UX designer and researcher with dual PhDs in HCI and Computing and 8+ years working on complex interfaces in healthcare, education, and AI. I both research and design: I lead formative studies and usability evaluations, then carry the findings directly into interface and interaction design and high-fidelity interactive prototypes. My focus areas are clinician workflows, error prevention, trust calibration in AI-assisted decisions, and accessibility. 15+ peer-reviewed publications at CHI, CSCW, and IUI. Comfortable working directly with engineers, clinicians, and regulatory stakeholders to turn usability findings into shipped design decisions.

CORE SKILLS

Human Factors · Usability Testing · Heuristic Evaluation · IEC 62366 · Clinical UX · Accessibility (WCAG 2.2) · Contextual Inquiry · Mixed Methods · Interactive Prototyping · Design Systems · Human–AI Interaction

EXPERIENCE

Human Factors & UX Designer

Aug 2026 – Present

iSono Health, Sunnyvale, CA

- Drive human factors research and interface design for ATUSA, a 3D breast ultrasound clinical viewer, translating radiologist workflow feedback into interface and interaction design across worklist, multi-view review, and 3D visualization tools.
- Conduct iterative usability evaluations with radiologists and clinical leads; translate findings into redesigns of the reading worklist, multi-view review panel, and 3D visualization environment.
- Design and build high-fidelity interactive clinical prototypes (React) covering linked cross-sectional navigation, measurement tools, and comparison views, used to align clinical, engineering, and product stakeholders.
- Establish interface conventions that visually separate AI-generated detections from human annotations, supporting trust calibration and reducing risk of misattributed findings.
- Author WCAG 2.2 AA accessibility conformance documentation and maintain traceability of design changes to requirements in support of regulatory readiness.

Senior Program Experience Researcher

Apr 2024 – Jul 2026

Western Governors University

- Led mixed-methods research to evaluate and improve program experiences for 400,000+ online learners, serving as sole researcher supporting multiple product and course teams.
- Conducted heuristic evaluations of course platforms and designed focus group protocols for studies of student retention and program dropout, working with cross-functional stakeholders from study design through readout.
- Ran competitive analyses benchmarking course experiences against other online learning providers and led interview studies on mobile learning, informing course design decisions.
- Translated findings into concrete design changes to course structure, assessment experience, and learning resources, presenting recommendations directly to product teams and leadership.

UX Research and Design Consultant

Sep 2025 – Jan 2026

iSono Health, Sunnyvale, CA

- Conducted usability research and design consultation for an AI-powered medical imaging device; delivered findings that supported clinical user workflow improvements.

Senior UX Researcher
Aug 2022 – Apr 2024
Western Governors University

- Designed and executed qualitative and quantitative studies, including interviews, surveys, and usability tests, across WGU's digital learning ecosystem.
- Contributed to research operations, helping to standardize study practices and make findings reusable across teams.
- Advanced accessibility and inclusive design for non-traditional learners, bringing their needs into course and platform decisions.
- Communicated findings to product and design teams to inform improvements across courses and learning resources.

Graduate Research Assistant
Aug 2019 – Jul 2024
Indiana University Indianapolis

- Led two NSF-funded projects on AI-assisted tools for older adults and visually impaired users; developed a design framework for balancing automation and user control, published at CHI and CSCW.
- Conducted 20+ in-depth interviews and contextual studies with older adults, caregivers, and blind/visually impaired participants; designed and evaluated the Scribe and TextFlow systems.

Research & Teaching Assistant
Jan 2016 – Jun 2019
University of North Carolina at Charlotte

- Designed and evaluated the Creative Sketching Partner, an AI co-creative design tool modeling conceptual shifts in sketching within NSF-funded research on human-AI collaboration; published at IUI 2020.
- Supported graduate instruction in Human-Computer Interaction, including course delivery, student mentorship, and assessment across HCI theory and research methods.

EDUCATION

Ph.D., Human-Computer Interaction *2019 – 2024*
Indiana University Indianapolis

Ph.D., Computing and Information Systems *2016 – 2019*
University of North Carolina at Charlotte

M.S., Telecommunications Engineering *2012 – 2015*
Politecnico di Milano

B.S., Electrical and Electronics Engineering *2006 – 2010*
Shiraz University

CERTIFICATIONS

- Human Factors & Usability Engineering, Arizona State University, 2026
- Patient Safety Specialization, Johns Hopkins University, 2026

SELECTED PUBLICATIONS

- **Karimi, P.**, & Martin-Hammond, A. (2025). Designing intelligent voice assistants for older adults' collaborative care: Exploring supportive and non-supportive interactions. *Proceedings of the ACM on Human-Computer Interaction (CSCW).*
- **Karimi, P.**, & Martin-Hammond, A. (2025). Finding the right balance: User control and automation in AI tools for supporting older adults' health information tasks. *CHI EA '25: Extended Abstracts of the CHI Conference on Human Factors in Computing Systems.*
- **Karimi, P.**, Bora, P., & Martin-Hammond, A. (2021). Scribe: Improving older adults' access to medical instructions from patient-physician conversations. *W4A '21: Proceedings of the 18th International Web for All Conference.*
- **Karimi, P.**, Rezwana, J., Siddiqui, S., Maher, M. L., & Dehbozorgi, N. (2020). Creative sketching partner: An analysis of human-AI co-creativity. *IUI '20: Proceedings of the 25th International Conference on Intelligent User Interfaces.*

15+ peer-reviewed publications at CHI, CSCW, and IUI. Full list on Google Scholar.